Linear Algebra and Its Applications

Section 6.3: Orthogonal Projections

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.3 Orthogonal projections

For an orthogonal basis $u_1, \ldots, u_p$ of W, the orthogonal component of y in W is

$$\widehat{y} = \operatorname{proj}_W y = \sum_{j=1}^{p} \frac{y \cdot u_j}{u_j \cdot u_j} u_j.$$

The residual $z = y - \widehat{y}$ lies in $W^{\perp}$. Thus $y = \widehat{y} + z$ splits a vector into a part in the subspace and a perpendicular part. If the columns of U are *orthonormal*, this simplifies to $\widehat{y} = UU^T y$. The formula without the denominators is generally wrong for merely orthogonal columns of nonunit length.

For any $w \in W$, $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2$. Therefore $\widehat{y}$ is the unique closest point and the distance is $\|z\|$. Each numerical calculation below checks the residual against every generator, not just against the projection.

Practice problem 6.3.P1: Recognize a vector already in the subspace

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 66 & 0 \\ 0 & 6 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{88}{66} = \frac{4}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-2}{6} = -\frac{1}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -7 & -1 \\ 1 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix} = \begin{bmatrix} -9 \\ 1 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The projection equals $y = (-9, 1, 6)^T$ itself; the residual is zero.

Exercise 6.3.1: Compute and check an orthogonal projection

Use the orthogonal basis formula for all four coefficients:

$$c_j = \frac{x \cdot u_j}{u_j \cdot u_j}, \qquad (c_1, c_2, c_3, c_4) = (-\frac{8}{9}, -\frac{2}{9}, \frac{2}{3}, 2).$$

The fourth component and the sum of the first three components are

$$q = c_4u_4 = \begin{bmatrix} 10 \\ -6 \\ -2 \\ 2 \end{bmatrix}, \qquad p = x - q = \begin{bmatrix} 0 \\ -2 \\ 4 \\ -2 \end{bmatrix}.$$

Here $p = c_1u_1 + c_2u_2 + c_3u_3$ is in the first requested span and q is in the second. The basis orthogonality makes $p \cdot q = 0$.

Conclusion. $x = p + q$ with the two explicit columns above.

Exercise 6.3.2: Compute and check an orthogonal projection

For the four orthogonal generators, compute $c_j = (v \cdot u_j)/(u_j \cdot u_j)$. The exact coefficients are

$$(c_1, c_2, c_3, c_4) = (2, \frac{3}{7}, \frac{12}{7}, -\frac{8}{7}).$$

Thus the requested split is

$$p = c_1u_1 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 2 \end{bmatrix}, \qquad q = c_2u_2 + c_3u_3 + c_4u_4 = \begin{bmatrix} 2 \\ 1 \\ -5 \\ 1 \end{bmatrix}.$$

Adding the two columns gives the original v. The first is in $\text{Span}\{u_1\}$ and the second in $\text{Span}\{u_2, u_3, u_4\}$, and they are orthogonal.

Conclusion. $v = p + q$ as displayed.

Exercise 6.3.3: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{3}{2} = \frac{3}{2}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{5}{2} = \frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{2} \\ \frac{5}{2} \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.4: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{25} = \frac{6}{5}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-15}{25} = -\frac{3}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & -4 \\ 4 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{6}{5} \\ -\frac{3}{5} \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.5: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 14 & 0 \\ 0 & 6 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{7}{14} = \frac{1}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-15}{6} = -\frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & 1 \\ -1 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ -\frac{5}{2} \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.6: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 18 & 0 \\ 0 & 2 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-27}{18} = -\frac{3}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{5}{2} = \frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -4 & 0 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{3}{2} \\ \frac{5}{2} \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.7: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 14 & 0 \\ 0 & 42 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{0}{14} = 0.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{28}{42} = \frac{2}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 3 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{2}{3} \\ \frac{8}{3} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{7}{3} \\ \frac{7}{3} \\ \frac{7}{3} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.8: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 \\ 0 & 14 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{3} = 2.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{7}{14} = \frac{1}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 \\ 1 & 3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{7}{2} \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{5}{2} \\ \frac{1}{2} \\ 2 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.9: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{3} = 2.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{10}{15} = \frac{2}{3}.$$
$$c_3 = \frac{y \cdot u_3}{u_3 \cdot u_3} = \frac{-2}{3} = -\frac{2}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ -1 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.10: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{1}{3} = \frac{1}{3}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{14}{3} = \frac{14}{3}.$$
$$c_3 = \frac{y \cdot u_3}{u_3 \cdot u_3} = \frac{-5}{3} = -\frac{5}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{14}{3} \\ -\frac{5}{3} \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -2 \\ 2 \\ 2 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.11: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 12 & 0 \\ 0 & 4 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{12} = \frac{1}{2}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{6}{4} = \frac{3}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & 1 \\ 1 & -1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 2 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.12: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 26 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{10} = 3.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{26}{26} = 1.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -4 \\ -2 & 1 \\ -1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.13: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 15 & 0 \\ 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{10}{15} = \frac{2}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-7}{3} = -\frac{7}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \\ -3 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ -\frac{7}{3} \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ -2 \\ 3 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ -4 \\ 4 \\ 0 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.14: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 14 & 0 \\ 0 & 49 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{7}{14} = \frac{1}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{0}{49} = 0.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 & 5 \\ 0 & -2 \\ -1 & 4 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ -\frac{3}{2} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 1 \\ 4 \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.15: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 35 & 0 \\ 0 & 14 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{35}{35} = 1.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-28}{14} = -2.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -3 & -3 \\ -5 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \\ -1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{40} = 2\sqrt{10}$, not the projection length.

Exercise 6.3.16: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 26 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{10} = 3.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{26}{26} = 1.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -4 \\ -2 & 1 \\ -1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{64} = 8$, not the projection length.

Exercise 6.3.17: Compute and check an orthogonal projection

(a) Compute the two distinct products.

$$U^T U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad UU^T = \begin{bmatrix} \frac{8}{9} & -\frac{2}{9} & \frac{2}{9} \\ -\frac{2}{9} & \frac{5}{9} & \frac{4}{9} \\ \frac{2}{9} & \frac{4}{9} & \frac{5}{9} \end{bmatrix}.$$

The first product checks orthonormal columns. The second acts on the ambient space and projects onto their span.

(b) Compute by the orthonormal projection formula.

$$\operatorname{proj}_W y = U(U^T y) = (UU^T)y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}.$$

Both methods are the same matrix product by associativity. Directly $U^T(y - \operatorname{proj}_W y) = 0$, checking the residual.

Exercise 6.3.18: Compute and check an orthogonal projection

(a) Compute the two distinct products.

$$U^T U = \begin{bmatrix} 1 \end{bmatrix}, \qquad UU^T = \begin{bmatrix} \frac{1}{10} & -\frac{3}{10} \\ -\frac{3}{10} & \frac{9}{10} \end{bmatrix}.$$

The first product checks orthonormal columns. The second acts on the ambient space and projects onto their span.

(b) Compute by the orthonormal projection formula.

$$\operatorname{proj}_W y = U(U^T y) = (UU^T)y = \begin{bmatrix} -2 \\ 6 \end{bmatrix}.$$

Both methods are the same matrix product by associativity. Directly $U^T(y - \operatorname{proj}_W y) = 0$, checking the residual.

Exercise 6.3.19: Compute and check an orthogonal projection

Plan. Remove the component of the given outside vector that lies in W; what remains is perpendicular to W.

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-2}{6} = -\frac{1}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{2}{30} = \frac{1}{15}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} \\ \frac{1}{15} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{2}{5} \\ \frac{4}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ \frac{2}{5} \\ \frac{1}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. Take $v = \begin{bmatrix} 0 \\ \frac{2}{5} \\ \frac{1}{5} \end{bmatrix}$. It is nonzero (its displayed entries are not all zero) and has zero dot product with u_1 and u_2.

Exercise 6.3.20: Compute and check an orthogonal projection

Plan. Remove the component of the given outside vector that lies in W; what remains is perpendicular to W.

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{1}{6} = \frac{1}{6}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-1}{30} = -\frac{1}{30}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{30} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{5} \\ -\frac{2}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ \frac{4}{5} \\ \frac{2}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. Take $v = \begin{bmatrix} 0 \\ \frac{4}{5} \\ \frac{2}{5} \end{bmatrix}$. It is nonzero (its displayed entries are not all zero) and has zero dot product with u_1 and u_2.

Exercise 6.3.21: Compute and check an orthogonal projection

(a) True. For any $w = c_1 u_1 + c_2 u_2$, $z \cdot w = c_1(z \cdot u_1) + c_2(z \cdot u_2) = 0$.

(b) True. This is the residual conclusion of the Orthogonal Decomposition Theorem.

(c) False. Orthogonal decomposition into W and $W^{\perp}$ is unique, so any orthogonal basis of the same W gives the same projection.

(d) True. For $y \in W$, the decomposition $y = y + 0$ already has its first part in W and its second in $W^{\perp}$, so uniqueness applies.

(e) True. With orthonormal columns, each projection coefficient is $u_j \cdot y$; collecting the coefficients gives $U(U^T y)$.

Exercise 6.3.22: Compute and check an orthogonal projection

(a) True. Such a vector is orthogonal to itself, so its squared length is zero and it must be zero.

(b) True. Each term $((y \cdot u_j)/(u_j \cdot u_j))u_j$ is the orthogonal projection onto the one-dimensional subspace $\text{Span}\{u_j\}$.

(c) True. The assumed sum is an orthogonal decomposition into W and $W^{\perp}$, and its uniqueness identifies the first part.

(d) False. The closest point is $\text{proj}_W y$; the stated vector is the residual. For a nonzero $y \in W$, the stated vector is zero while the unique best approximation is y.

(e) False. UU^T fixes only the vectors in $\text{Col}\, U$ in general. For $U = (1, 0)^T$ and $x = (0, 1)^T$, $UU^T x = 0 \neq x$.

Exercise 6.3.23: Compute and check an orthogonal projection

Apply orthogonal decomposition to the row space.

Theorem 3 identifies $(\text{Row}\, A)^{\perp} = \text{Nul}\, A$. Taking $W = \text{Row}\, A$ in the decomposition theorem gives $x = p + u$ with $p \in \text{Row}\, A$ and $u \in \text{Nul}\, A$.

Existence of a row-space solution for a consistent equation.

If $Ax = b$ has a solution x, decompose that solution as above. Since $Au = 0$, we obtain $Ap = A(x - u) = b$. Thus some row-space vector does solve the equation.

Uniqueness within the row space.

If p, p' are row-space solutions, then $A(p - p') = 0$. Their difference lies both in the row space and its perpendicular null space, so it is zero. Therefore $p = p'$.

Conclusion. Every vector has the required split, and every consistent system has exactly one solution belonging to the row space.

Exercise 6.3.24: Compute and check an orthogonal projection

(a) Check every type of pair.

Two distinct w_i are orthogonal by assumption, as are two distinct v_j. A mixed pair is orthogonal because $w_i \in W$ and $v_j \in W^\perp$. Thus the concatenated family is orthogonal. Its members are all nonzero because they belong to bases.

(b) Span an arbitrary ambient vector.

Every $x \in \mathbb{R}^n$ splits as $x = w + v$, where $w \in W$ and $v \in W^\perp$. Expand w in the w_i basis and v in the v_j basis. Adding the expansions expresses x using the concatenated family.

(c) Count the resulting basis.

A nonzero orthogonal family is independent, and (b) proves spanning. Hence this family is a basis of $\mathbb{R}^n$ with $p + q$ members.

Conclusion. $\dim W + \dim W^\perp = p + q = n$. This includes the zero-space boundary cases, where one of the bases is empty.

Exercise 6.3.25: Compute and check an orthogonal projection

Use the normalized matrix from Exercise 6.2.36.

That matrix is $U = A/10$, with the explicitly supplied 8×4 integer matrix A from Exercise 6.2.35. It satisfies $U^T U = I_4$.

$$\widehat{y} = U(U^T y) = \begin{bmatrix} \frac{6}{5} \\ \frac{3}{5} \\ \frac{1}{5} \\ \frac{2}{5} \\ \frac{6}{5} \\ \frac{3}{5} \\ \frac{1}{5} \\ \frac{2}{5} \end{bmatrix}, \qquad y - \widehat{y} = \begin{bmatrix} -\frac{1}{5} \\ \frac{3}{5} \\ -\frac{1}{5} \\ -\frac{1}{5} \\ \frac{3}{5} \\ -\frac{1}{5} \\ \frac{3}{5} \\ \frac{3}{5} \end{bmatrix}.$$

Commands reproducing the exact calculation.

```
from sympy import Matrix, ones
A = Matrix([[-6,-3,6,1],[-1,2,1,-6],[3,6,3,-2],
            [6,-3,6,-1],[2,-1,2,3],[-3,6,3,2],
            [-2,-1,2,-3],[1,2,1,6]])
U = A/10
y = ones(8,1)
p = U*(U.T*y)
print(p)
assert U.T*(y-p) == Matrix.zeros(4,1)
```

The zero dot-product column $U^T(y - p) = 0$ checks that the residual is perpendicular to the subspace, proving that the displayed point is its projection and hence the unique closest point.

Exercise 6.3.26: Compute and check an orthogonal projection

Use the normalized matrix from Exercise 6.2.36.

That matrix is $U = A/10$, with the explicitly supplied 8×4 integer matrix A from Exercise 6.2.35. It satisfies $U^T U = I_4$.

$$\widehat{y} = U(U^T y) = \begin{bmatrix} \frac{1}{5} \\ \frac{23}{25} \\ \frac{11}{25} \\ 1 \\ -\frac{1}{5} \\ -\frac{11}{25} \\ \frac{3}{5} \\ -\frac{23}{25} \end{bmatrix}, \qquad y - \widehat{y} = \begin{bmatrix} \frac{4}{5} \\ \frac{2}{25} \\ \frac{14}{25} \\ 0 \\ -\frac{4}{5} \\ -\frac{14}{25} \\ -\frac{8}{5} \\ -\frac{2}{25} \end{bmatrix}.$$

Conclusion. The distance is $\|y - \widehat{y}\| = \sqrt{\frac{112}{25}} = \frac{4\sqrt{7}}{5}$. Here y denotes the exercise target b.